

Thermodynamic State Vector Interface Contracts and Monad Execution in the Web of Life Engine

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Sprint 23 Academic Preprint

Abstract

This preprint details the architectural formalization and implementation of thermodynamic state vector interface contracts within the Web of Life engine (`src/thermodynamics/types.ts` and `src/thermodynamics/thermodynamic_monad_process.ts`). Grounded in non-equilibrium thermodynamics and systems ecology, we establish strict runtime enforcement of the First Law of Thermodynamics (energy conservation) and the Second Law of Thermodynamics (entropy generation rate $\dot{S}_{\text{gen}} \geq 0$ and the Gouy-Stodola exergy destruction theorem $\dot{I} = T_0 \dot{S}_{\text{gen}}$). These contracts ensure absolute biophysical fidelity for planetary-scale ecological simulations.

1 Introduction and Systems Ecology Context

The Web of Life simulation engine models planetary ecosystems as open thermodynamic control volumes driven by solar exergy input and bounded by thermal radiation to deep space. To prevent unphysical energy creation or entropy destruction, Sprint 23 formalizes thermodynamic state vectors into rigorous TypeScript interface contracts and executable monad transformations.

2 Governing Thermodynamic Equations

2.1 First Law of Thermodynamics (Energy Conservation)

For the global Earth Pod domain under closed-mass/open-energy planetary boundary conditions, internal energy evolution is governed by:

$$\frac{dU_{\text{Earth}}}{dt} = \dot{\Phi}_{\text{solar}} - \dot{\Phi}_{\text{thermal}} + \dot{W}_{\text{boundary}} \quad (1)$$

2.2 Second Law of Thermodynamics (Entropy Balance)

The entropy rate balance for the control volume is expressed as:

$$\frac{dS}{dt} = \sum_j \frac{\dot{Q}_j}{T_j} + \sum_k \dot{m}_k s_k + \dot{S}_{\text{gen}} \quad (2)$$

where internal entropy generation satisfies the Clausius inequality for irreversible processes:

$$\dot{S}_{\text{gen}} \geq 0 \quad (3)$$

2.3 Exergy Destruction Rate ($\dot{I}$)

By the Gouy-Stodola Theorem, exergy destruction relative to ambient reference temperature $T_0 = 288.15$ K is strictly linked to entropy generation:

$$\dot{I} = T_0 \dot{S}_{\text{gen}} \geq 0 \quad (4)$$

3 Interface Contracts & Monad Architecture

The core types are structured to compose boundary fluxes and exergy metrics immutably:

- **IBoundaryFlux**: Tracks energy and mass transfer rates with associated boundary temperatures.
- **IExergyDestructionMetrics**: Encapsulates T_0 , $\dot{S}_{\text{gen}}$, $\dot{I}$, input exergy, and exergetic efficiency.
- **IThermodynamicStateVector**: Combines internal energy, total entropy, flux structures, and exergy metrics.
- **ThermodynamicMonadProcess**: Enforces invariant checks ($\dot{S}_{\text{gen}} \geq 0$ and net mass balance $= 0$) on every state transition step.

4 Verification and Test Results

Unit tests established in `tests/sprint_023.test.ts` confirm that any state vector violating Second Law invariants or exhibiting negative entropy generation throws explicit thermodynamic violation exceptions. Valid solar and terrestrial flux steps successfully maintain energy conservation and non-negative exergy destruction rates.

5 Conclusion

Sprint 23 provides a mathematically rigorous foundation for thermodynamic modeling in the Web of Life repository. Future work will leverage these state vectors to examine ecosystem succession and thermodynamic efficiency under varying climatic forcings.

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